Numerical equivalence, cohomological equivalence, and Lefschetz theory for abelian varieties over finite fields

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To Jean-Pierre Serre, in testimony of our admiration

Abstract. We determine the group of automorphisms generated, inside the endomorphisms of the cohomology of an abelian variety over a finite field, by the Lefschetz operators and the complex multiplications.

Keywords. Algebraic geometry (abelian varieties, étale cohomology); 14F20, 14K22.

1. Let k be an algebraic closure of the field $\mathbb{F}_p$ with p elements. We write ℓ for a prime number different from p. We fix, over k, an isomorphism $\mathbb{Q}_\ell \simeq \mathbb{Q}_\ell(1)$.

Let A be an abelian variety over k. Write Z, or $Z(A)$, for the group of cycles with rational coefficients on A, graded by codimension:

$$Z = \bigoplus_i Z^i.$$

On Z^i one has numerical equivalence and cohomological equivalence, for étale cohomology with coefficients in $\mathbb{Q}_\ell$. Cohomologically equivalent cycles are numerically equivalent.

Let A_j be the simple factors of A, let E_j be the centre of $\mathrm{End}(A_j) \otimes \mathbb{Q}$, and let F be the compositum, in an algebraic closure of $\mathbb{Q}$, of the Galois closures of the E_j. Thus F is either a totally real field or a CM field. If ℓ is a prime, let $D_\ell \subset \mathrm{Gal}(F/\mathbb{Q})$ be a decomposition group at ℓ; let $c \in \mathrm{Gal}(F/\mathbb{Q})$ be complex conjugation.

Theorem 1. *If $c \in D_\ell$, then numerical equivalence and cohomological equivalence, for ℓ-adic étale cohomology, coincide for A.*

A less precise theorem, in which the condition on ℓ depended, for a simple variety, on the choice of a maximal field of complex multiplication, was proved in [2].

The following remarks are evident:

(a) Once A is fixed, the condition of Theorem 1 is satisfied, by the Chebotarev theorem, for a set L of primes ℓ of non-zero analytic or naive density.

(b) If the condition of Theorem 1 is satisfied for A, it is also satisfied for the powers A^n of A. Hence there is a set L, of density > 0, such that the conclusion holds for every power A^n. This was observed by Milne [5].

Put

$$H = \bigoplus_{i=0}^{2g} H^i(A, \mathbb{Q}_\ell) = \bigoplus_{i=0}^{2g} H^i, \qquad g = \dim(A),$$

and put $D = \mathrm{End}(A) \otimes \mathbb{Q}$. The group $D^\times$ acts on H, as does the group, isomorphic to $\mathrm{SL}(2, \mathbb{Q})$, supplied by Lefschetz theory. Both preserve algebraic cohomology classes. Let $G \subset \mathrm{GL}(H)$ be the Zariski closure of the group generated by $D^\times$ and $\mathrm{SL}(2)$. The key point in our proof of Theorem 1 is a description of G, or rather of the group $G_{\mathbb{Q}_\ell}$ over $\mathbb{Q}_\ell$ obtained from it by extension of scalars. In §4 we shall show how this description reduces to the following two cases.

[0]Written from a note of Deligne and a letter of 12 July 1997.

First let E be a supersingular elliptic curve and let $A = E^n$. The group $\mathrm{End}(A)^\times$, acting on

$$H \otimes \mathbb{Q}_\ell = \Lambda^\bullet(H^1 \otimes \mathbb{Q}_\ell),$$

has Zariski closure $\mathrm{GL}(H^1 \otimes \mathbb{Q}_\ell)$. Let $V = H^1 \otimes \mathbb{Q}_\ell$ and let V^* be its dual.

Theorem 2. *The group $G_{\mathbb{Q}_\ell}$ is isomorphic to the group $\mathrm{CSpin}(V \oplus V^*)$ of spin similitudes, where $V \oplus V^*$ is endowed with the natural quadratic form. It acts on*

$$H \otimes \mathbb{Q}_\ell = \bigoplus_i H^i(A, \mathbb{Q}_\ell)$$

by the sum of its two half-spin representations. In particular, the representation on even-degree classes, respectively odd-degree classes, is irreducible.

At the other extreme, let A be an irreducible variety such that $\mathrm{End}(A) \otimes \mathbb{Q} = D$ is a division algebra, of reduced degree n, over an imaginary quadratic field E; A has dimension n.

Let $c \in \mathrm{Gal}(E/\mathbb{Q})$ be complex conjugation, and choose an embedding j of E into $\mathbb{Q}_\ell$. The E-module structure of $H^1(A, \mathbb{Q}_\ell)$ gives a decomposition

$$H^1 = H_1' \oplus H_1'',$$

where $\lambda \in E$ acts on H_1', respectively on H_1'', by multiplication by $j(\lambda)$, respectively by $j(c\lambda)$. Let $V = V_1 \oplus V_2$ be the dual decomposition of $H^1(A, \mathbb{Q}_\ell)$. The Zariski closure of $D^\times$ becomes, over $\mathbb{Q}_\ell$, the group $\mathrm{GL}(V_1) \times \mathrm{GL}(V_2)$.

Endow

$$H^\bullet(A, \mathbb{Q}_\ell) = \bigoplus_{p,q} \Lambda^p V_1 \otimes \Lambda^q V_2$$

with the grading given by $k := p - q$:

$$\mathrm{gr}_k = \bigoplus_{p-q=k} \Lambda^p V_1 \otimes \Lambda^q V_2. \tag{1.1}$$

The part of degree $-n$ is therefore $\Lambda^0 V_1 \otimes \Lambda^n V_2$, of dimension 1, and the part of degree $-n+1$ is

$$V_1 \otimes \Lambda^n V_2 \oplus \Lambda^0 V_1 \otimes \Lambda^{n-1} V_2,$$

of dimension $2n$.

Theorem 3. *(i) For an isomorphism from $\bigoplus \Lambda^p V_1 \otimes \Lambda^q V_2$ to $\Lambda^\bullet(\mathrm{gr}_{-n+1})$ carrying gr_k to $\Lambda^{n+k}\,\mathrm{gr}_{-n+1}$, $G_{\mathbb{Q}_\ell}$ identifies with the group generated by $\mathrm{GL}(\mathrm{gr}_{-n+1})$ acting on $\Lambda^\bullet\,\mathrm{gr}_{-n+1}$ and by the homotheties $x \mapsto \lambda x$.*

(ii) The gr_k $(-n \le k \le n)$ are therefore irreducible, pairwise non-isomorphic representations of G. The subspaces gr_k and gr_ℓ are orthogonal for the cup product if $k + \ell \neq 0$, and are in perfect duality if $k + \ell = 0$.

2. We explain how to reduce Theorem 1, and even a slightly stronger variant, cf. §5, to the analysis of the group G.

(a) Let Z_0 be the $\mathbb{Q}$-vector subspace of $H^\bullet(A, \mathbb{Q}_\ell)$ which is the image of Z, and let Z_1 be the $\mathbb{Q}_\ell$-vector subspace generated by Z_0. We shall show that the intersection form is non-degenerate on Z_1. Its restriction, which is rational, to Z_0 therefore has zero kernel: it is non-degenerate and induces a surjection from Z_1 to $\mathrm{Hom}(Z_0, \mathbb{Q}_\ell)$. Hence

$$\dim_{\mathbb{Q}_\ell}(Z_1) \geq \dim_{\mathbb{Q}}(Z_0),$$

the elements of a basis of Z_0 are linearly independent over $\mathbb{Q}_\ell$, and $Z_0 \otimes \mathbb{Q}_\ell \simeq Z_1$.

(b) Let $\mathbb{Q}_\lambda$ denote a sufficiently large finite Galois extension of $\mathbb{Q}_\ell$. It is then enough to check the non-degeneracy of the intersection form on

$$Z_\lambda := Z_1 \otimes_{\mathbb{Q}_\ell} \mathbb{Q}_\lambda \subset H_\lambda := H^\bullet(A, \mathbb{Q}_\lambda).$$

(c) Let L be the Lefschetz operator on H obtained from a polarization of A, and let $\Lambda : H^i \to H^{i-2}$ be the associated operator [4]. The exponentials of the nilpotent endomorphisms L and Λ of H generate a subgroup S, isomorphic to $\mathrm{SL}(2, \mathbb{Q})$. Let G be the Zariski closure, in $\mathrm{GL}(H)$, of the subgroup generated by S and by $(\mathrm{End}(A) \otimes \mathbb{Q})^\times$. It preserves Z_1, hence Z_λ; the fact that Λ preserves algebraic cycles is due to Lieberman, cf. [4]. Since G contains the group $\mathbb{G}_m$ corresponding to the grading of $H^\bullet(A)$, via the action of $(\mathbb{Q} \cdot 1)^\times \subset \mathrm{End}(A) \otimes \mathbb{Q}$, and also the group corresponding to the same grading shifted by $\dim(A)$, via the diagonal matrices in S, it contains the homotheties $x \mapsto \lambda x$.

(d) Assume that $\mathbb{Q}_\lambda$ is a sufficiently large Galois extension of $\mathbb{Q}_\ell$. We shall check that H_λ is a sum of absolutely irreducible, pairwise non-isomorphic representations W of G, and that the intersection form defines non-degenerate pairings on $W \times W'$, where $W \mapsto W'$ is an involution of the factors. Assume that there exists $\sigma \in \mathrm{Gal}(\mathbb{Q}_\lambda/\mathbb{Q}_\ell)$ which exchanges the dual factors. Since Z_λ is stable under the Galois group, the intersection form is then non-degenerate; this will finish the proof.

Let E be the centre of $\mathrm{End}(A) \otimes \mathbb{Q}$: E is a product of CM fields E_i; we agree to count a totally real field as CM. Assume that E has the following property, equivalent to the hypothesis $c \in D_\ell$ of Theorem 1:

there are a Galois extension $\mathbb{Q}_\lambda/\mathbb{Q}_\ell$ and $\sigma \in \mathrm{Gal}(\mathbb{Q}_\lambda/\mathbb{Q}_\ell)$ such that, for all i and every embedding $\tau : E_i \to \mathbb{Q}$ (2.1)

where c is complex conjugation on E_i. We shall then verify below the existence of σ exchanging W and W'. Under this hypothesis, Theorem 1 follows.

3. We recall some linear algebra, referring to [3], based on [1]. Let V be a vector space over a field K (here $\mathbb{Q}_\ell$, $\mathbb{Q}_\lambda$, or $\overline{\mathbb{Q}}_\ell$), let V^* be its dual, and put on $W = V \oplus V^*$ the non-degenerate quadratic form

$$Q(v + \lambda) = \langle \lambda, v \rangle \qquad (v \in V,\ \lambda \in V^*).$$

Let C be the associated Clifford algebra, the quotient of the tensor algebra $T(V \oplus V^*)$ by the ideal generated by the elements $x \cdot x - Q(x)$, $x \in V \oplus V^*$. It is a $\mathbb{Z}/2\mathbb{Z}$-graded algebra [3]; write this grading as $C = C^+ \oplus C^-$. Let W act on the exterior algebra $\Lambda^\bullet V$ of V by

$$\begin{aligned} v \cdot x &= v \wedge x && (v \in V,\ x \in \Lambda^\bullet V), \\ \lambda \cdot x &= \lambda \perp x = i(\lambda)x && (\lambda \in V^*,\ x \in \Lambda^\bullet V). \end{aligned} \tag{3.1}$$

The defining relation of C is satisfied, and (3.1) therefore extends to a C-module structure

$$C \longrightarrow \mathrm{End}(\Lambda^\bullet V). \tag{3.2}$$

The morphism (3.2) is an isomorphism: see [1], Thm. 2; alternatively, observe that if $V = V' \oplus V''$, then C is the tensor product, in the $\mathbb{Z}/2$-graded sense, of the analogous algebras C' and C'', reducing one to the easy case where V has rank 1. The image of C^+ under (3.2) is the algebra of endomorphisms of $\Lambda^\bullet V$ respecting the $\mathbb{Z}/2$-grading of $\Lambda^\bullet V$.

The group $C^\times$ acts on C by conjugation. The subgroup H normalizing $W \subset C$ respects the $\mathbb{Z}/2$-grading of C, and hence consists of even or odd elements. Its action on W defines $H \to \mathrm{O}(W)$, whose kernel is the centre of $C^\times$. The subgroup $(C^+)^\times \cap H$ of $(C^+)^\times$ which normalizes W is the group of spin similitudes $\mathrm{CSpin}(W)$. It maps to $\mathrm{SO}(W)$:

$$1 \longrightarrow \mathbb{G}_m \longrightarrow \mathrm{CSpin}(W) \longrightarrow \mathrm{SO}(W) \longrightarrow 1.$$

Thus $\mathrm{CSpin}(W)$ is the group of automorphisms of $\Lambda^\bullet V$ respecting the $\mathbb{Z}/2$-grading and the subspace W of $\mathrm{End}(\Lambda^\bullet V)$. Its Lie algebra is the algebra of even endomorphisms T of $\Lambda^\bullet V$ such that, with W identified with a subspace of $\mathrm{End}(\Lambda^\bullet V)$, one has $[T, W] \subset W$. The representation $\Lambda^\bullet V$ of $\mathrm{CSpin}(W)$ is the sum of the two half-spin representations.

It is clear that $\mathrm{GL}(V)$, acting on $\Lambda^\bullet V$, is a subgroup of $\mathrm{CSpin}(W)$. In CSpin it lifts the Levi subgroup of $\mathrm{SO}(W)$, the intersection of the opposite parabolic subgroups preserving the maximal isotropic subspaces V and V^* of W.

The maximal isotropic subspace $V^* \subset W \subset C$ and the line $K \cdot 1$ in $\Lambda^\bullet V$ determine one another: V^* is the subspace of W annihilating 1, and $K \cdot 1$ is the subspace of $\Lambda^\bullet V$ annihilated by V^*. By the same rule, V corresponds to the line $\Lambda^{\max} V$.

The same story repeats for any maximal isotropic subspace of W. The case of interest to us is the case where $V = V_1 \oplus V_2$, and where W is decomposed as the sum of two maximal isotropic subspaces by

$$W = (V_1 \oplus V_2^*) \oplus (V_1^* \oplus V_2).$$

The annihilator of $V_1^* \oplus V_2$ is the line $\Lambda^{\max} V_2 \subset \Lambda^\bullet V$. Fix a generator Ω of it. The inclusion in W of the complementary isotropic subspace $V_1 \oplus V_2^*$ extends to a morphism of algebras

$$\Lambda^\bullet(V_1 \oplus V_2^*) \longrightarrow C,$$

and

$$I : \Lambda^\bullet(V_1 \oplus V_2^*) \longrightarrow \Lambda^\bullet V, \qquad \lambda \longmapsto \lambda\Omega \tag{3.3}$$

is an isomorphism. The super-commutator with $x \in W$ stabilizes $\Lambda^\bullet(V_1 \oplus V_2^*) \subset C$ and induces a super-derivation. For $x \in V_1^* \oplus V_2$, identified with the dual of $V_1 \oplus V_2^*$, this derivation is interior product $x \perp$. Through the isomorphism I, the action of $v \in V_1 \oplus V_2^* \subset W$ therefore becomes exterior product, while that of $x \in V_1^* \oplus V_2 \subset W$, which annihilates Ω, becomes interior product.

The exterior product in $\Lambda^\bullet(V_1 \oplus V_2^*)$, transported by I, defines a new product, denoted $*$, on $\Lambda^\bullet V$. If $\Lambda^\bullet(V_1 \oplus V_2^*)$ is graded by putting V_1 in degree 1 and V_2^* in degree -1, the product is compatible with the grading. The isomorphism I transports it to the evident grading of $\Lambda^\bullet V$, shifted by $\dim(V_2)$ so as to put Ω in degree 0. Thus the product $*$ is compatible with the shifted evident grading. The natural grading of $\Lambda^\bullet(V_1 \oplus V_2^*)$, for which V_2^* has degree 1, is also compatible with the product. One then obtains on $\Lambda^\bullet V$ the grading gr_k of (1.1), shifted by $\dim(V_2)$; $1 \in \Lambda^\bullet V_2$, of degree 0 for gr_k, has degree $\dim(V_2)$.

For $v_1 \in V_1$ and $v_2 \in V_2$, compute

$$(v_1 \wedge v_2)\wedge : \Lambda^\bullet V \longrightarrow \Lambda^\bullet V$$

using the isomorphism (3.3). One has the diagram

$$\begin{tikzcd}
\Lambda^\bullet(V_1 \oplus V_2^*) \arrow[r, "I"] \arrow[d, equal] & \Lambda^\bullet V \arrow[d, equal] \\
\Lambda^\bullet V_1 \otimes \Lambda^\bullet V_2^* \arrow[r] & \Lambda^\bullet V_1 \otimes \Lambda^\bullet V_2.
\end{tikzcd}$$

and the bottom map is a tensor product of maps. If $a \in \Lambda^\bullet V_1$ and $b \in \Lambda^\bullet V_2$, then

$$(v_1 \wedge v_2)(a \wedge b) = (-1)^{\deg a}(v_1 \wedge a) \wedge (v_2 \wedge b).$$

For the isomorphism $\Lambda^\bullet V_2^* \to \Lambda^\bullet V_2$ defining I, the map $b \mapsto v_2 \wedge b$ is transported to $b^* \mapsto v_2 \perp b^*$. It follows that $(v_1 \wedge v_2)\wedge$ is transported to the endomorphism $v_1 \wedge v_2 \perp$ of $\Lambda^\bullet(V_1 \oplus V_2^*)$, which is immediately checked to be a derivation. This derivation preserves $V_1 \oplus V_2^*$.

4. Computing the group G of §1, over a sufficiently large extension of $\mathbb{Q}_\ell$, is amusing. Here is the story.

(a) Let E be the centre of $D = \mathrm{End}(A) \otimes \mathbb{Q}$, a product of CM fields E_i; D is central simple over E.

(b) After extending scalars from $\mathbb{Q}$ to a sufficiently large $\mathbb{Q}_\lambda$, E becomes a sum of copies of $\mathbb{Q}_\lambda$, indexed by the set Σ of morphisms of $\mathbb{Q}$-algebras $E \to \mathbb{Q}_\lambda$. After extending scalars from $\mathbb{Q}_\ell$ to $\mathbb{Q}_\lambda$, $H^1(A)$ becomes $\bigoplus H_\sigma$, $\sigma \in \Sigma$, where

$$H_\sigma = \{\alpha \in H^1(A, \mathbb{Q}_\lambda) : [z]\alpha = \sigma(z)\alpha \ (z \in E)\},$$

where $[z]$ denotes the action of E on $H^1(A)$ obtained from its action on A. The algebra $D \otimes \mathbb{Q}_\lambda$ is

$$\prod_\sigma \mathrm{End}(H_\sigma).$$

The group G, acting on $\Lambda^\bullet(\bigoplus_\sigma H_\sigma)$, is generated by the product of the $\mathrm{GL}(H_\sigma)$ and by a "diagonal" $\mathrm{SL}(2)$. Let

$$L : H^i(A, \mathbb{Q}_\lambda) \longrightarrow H^{i+2}(A, \mathbb{Q}_\lambda)$$

be the Lefschetz operator, identified with a cohomology class in

$$H^2(A, \mathbb{Q}_\lambda) = \Lambda^2\Big(\bigoplus_\sigma H_\sigma\Big).$$

Lemma 2.1 of [2] implies that

$$L = \sum_v L_v,$$

where v runs through the quotient set $\Sigma/\{1, c\}$, complex conjugation acting on Σ through its action on E, and where

$$L_v \in \Lambda^2(H_\sigma \oplus H_{c\sigma})$$

for $v = \{\sigma, c\sigma\}$. It may happen that σ is fixed by c, in which case $L_v \in \Lambda^2 H_\sigma$. Each L_v lies in the closure of the orbit of L under $\prod \mathrm{GL}(H_\sigma)$, and hence in the Lie algebra of G.

The same arguments give an analogous decomposition of the operator Λ; the commutation relations are preserved by the components L_v and Λ_v. Consequently, G is also generated by the product of the $\mathrm{GL}(H_\sigma)$ and by one copy of $\mathrm{SL}(2)$, defined over $\mathbb{Q}_\lambda$, for each orbit of c in Σ. Thus G is a product acting on a tensor product.

(c) Story for $\{\sigma\}$, $\sigma = c\sigma$. This corresponds to a possible factor $\mathbb{Q}$ of E, hence to a factor of A isogenous to $B = E^n$, where E is a supersingular elliptic curve over k.

Let $V = H_\sigma = H^1(B, \mathbb{Q}_\lambda)$, a vector space of even dimension. Let $W = V \oplus V^*$. The Lefschetz class $L \in \Lambda^2 V$ is an isomorphism from V to V^*. We shall show that the group generated by $\mathrm{GL}(V)$ and by the subgroup $\mathrm{SL}(2)$ associated with L is $\mathrm{CSpin}(W)$ in its spin representation on $H^\bullet(B) = \Lambda^\bullet V$.

It is immediate that, for its action on $\Lambda^\bullet V$ by left multiplication, L satisfies

$$[L, W] \subset W.$$

Therefore L is an element of Lie(CSpin(W)); it lies in the Lie algebra of the unipotent radical N of the Levi subgroup defined by $W = V \oplus V^*$ (§3), and is a non-singular element of it. The conjugates, by $\mathrm{GL}(V)$, of the exponential of L therefore generate N. Similarly, Λ and its conjugates generate the Lie algebra of the opposite unipotent subgroup. Consequently the group generated by $\mathrm{GL}(V)$ and $\mathrm{SL}(2)$ contains the Spin group; we have already seen that it contains the homotheties (§2, (c)). Its Zariski closure, over $\mathbb{Q}_\ell$, is therefore $\mathrm{CSpin}(V \oplus V^*)$, proving Theorem 2.

(d) Story for $\{\sigma, c\sigma\}$, $\sigma \neq c\sigma$. In this case

$$H_v = H_\sigma \oplus H_{c\sigma} = V_1 \oplus V_2,$$

and $(\mathrm{End}(A) \otimes \mathbb{Q}_\lambda)^\times$ acts on H_v by $\mathrm{GL}(V_1) \times \mathrm{GL}(V_2)$.

Write, as in §3,

$$\Lambda^\bullet(V_1 \oplus V_2) \xrightarrow[I^{-1}]{\simeq} \Lambda^\bullet(V_1 \oplus V_2^*).$$

Endow $\Lambda^\bullet(V_1 \oplus V_2)$ with the product $*$, which under I^{-1} becomes the exterior product on the right. One can write ([2, Lemma 2.1])

$$L = \sum e_i \wedge f_i \in \Lambda^2(V_1 \oplus V_2),$$

with $e_i \in V_1$ and $f_i \in V_2$. The isomorphism I^{-1} exchanges the action of L on $\Lambda^\bullet(V_1 \oplus V_2)$ with the action on the right-hand side given by

$$\sum e_i \wedge f_i \perp .$$

This is a derivation (§3); in fact, in the bases of V_1 and V_2^* given by (e_i, f_i^*), it is the sum of the endomorphisms

$$x = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{sl}(2)$$

of k^2, hence the corresponding endomorphism of k^{2n} extended, as a Lie-algebra action, to

$$\Lambda^\bullet(V_1 \oplus V_2^*) = \Lambda^\bullet(k^{2n}).$$

Thus the action of L on $\Lambda^\bullet(V_1 \oplus V_2)$ leaves the product $*$ invariant, in the Lie-algebra sense; it is therefore a derivation.

The element

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

of $\mathfrak{sl}(2)$ acts on $\Lambda^i V$ by $i - n = \mathrm{gr}_1(\Lambda^i V)$, where gr_1 is the shifted natural grading (§3). Since this grading is compatible with the product $*$, the product is invariant under the Borel subalgebra of $\mathfrak{sl}(2)$. But in a representation of $\mathfrak{sl}(2)$, a vector killed by a Borel subalgebra is killed by all of $\mathfrak{sl}(2)$. It follows that $*$ is invariant under $\mathrm{SL}(2)$.

Consider the decomposition of $\Lambda^\bullet V$ given by (1.1):

$$\mathrm{gr}_k \Lambda^\bullet V = \bigoplus_{p-q=k} \Lambda^p V_1 \otimes \Lambda^q V_2.$$

This decomposition is invariant under L and Λ. Indeed, E, through its embedding in $\mathbb{Q}_\lambda$, acts on $\Lambda^\bullet V$. Let $U(1)$ be the group

$$\{\lambda \in E : N_{E/F}\lambda = 1\},$$

where F is the subalgebra of E formed by the fixed points of c. If $\lambda \in U(1)$, then λ acts on gr_k by the scalar $\lambda^{p-q} \in \mathbb{Q}_\lambda$; it commutes with L and Λ, and the λ^{p-q} are distinct characters of $U(1)$. Summarizing:

Lemma 4.1. *Endow $\Lambda^\bullet V$ with the product $*$ and the grading (1.1).*

(i) $\Lambda^\bullet V$ is generated by

$$\mathrm{gr}_{1-n} = V_1 \otimes \Lambda^n V_2 \oplus \Lambda^0 V_1 \otimes \Lambda^{n-1} V_2.$$

(ii) The product $* : \mathrm{gr}_i \times \mathrm{gr}_j \to \mathrm{gr}_{i+j}$ is invariant under $\mathrm{SL}(2)$, $\mathrm{SL}(V_1)$, *and* $\mathrm{SL}(V_2)$.

(iii) The grading of $\Lambda^\bullet V$ is preserved by G.

The image of G in $\mathrm{GL}(\mathrm{gr}_{n-1})$ contains $\mathrm{GL}(V_1)$, $\mathrm{GL}(V_2) \simeq \mathrm{GL}(\Lambda^{n-1} V_2)$, as well as an isomorphism $\Lambda^0 V_1 \otimes \Lambda^n V_2 \to V_1 \otimes \Lambda^n V_2$ supplied by L and an inverse isomorphism supplied by Λ. It is therefore $\mathrm{GL}(\mathrm{gr}_{n-1})$. Property (i) identifies $\Lambda^\bullet V$ with $\Lambda^\bullet(\mathrm{gr}_{n-1})$, compatibly with the action of $\mathrm{SL}(2) \times \mathrm{SL}(V_1) \times \mathrm{SL}(V_2)$. The group generated by $\{zs_1, z^{-1}s_2\}$, where z is a scalar and $s_i \in \mathrm{SL}(V_i)$, in $\mathrm{GL}(V_1) \times \mathrm{GL}(V_2)$ acts on $\Lambda^\bullet V$ by the action of $\mathrm{GL}(\mathrm{gr}_{1-n})$ described in Theorem 3. Thus G is generated by $\mathrm{GL}(\mathrm{gr}_{1-n})$, isomorphic to $\mathrm{GL}(2n)$, and by a one-dimensional torus, acting by global homotheties on $\Lambda^\bullet V$. This proves Theorem 3 (i). It is clear that, already for the action of $\mathrm{GL}(\mathrm{gr}_{1-n})$, the representations obtained are absolutely irreducible and pairwise non-isomorphic. Finally, if $k + \ell \neq 0$, the spaces gr_k and gr_ℓ are not dual representations, even for the action of $\mathrm{GL}(2n) \subset G$. Since the cup product is non-degenerate, gr_k and gr_ℓ are therefore in perfect duality.

5. Once the description of G is known, the proof of Theorem 1 is as follows. Write

$$H_\lambda = H^\bullet(A, \mathbb{Q}_\lambda) = \bigotimes_{v \in \Sigma/c} H_v^\bullet,$$

where $H_v^\bullet$ is as in §4. Assume $\mathbb{Q}_\lambda$ sufficiently large and Galois over $\mathbb{Q}_\ell$. Under the action of the product, now denoted G, of the groups described in §4, H_λ decomposes as a direct sum indexed by functions $\sigma \mapsto k(\sigma)$, $\sigma \in \Sigma$, where

$$k(\sigma) \in \mathbb{Z} \qquad \text{if } \sigma \neq \sigma c; \quad \text{then } k(\sigma c) = -k(\sigma), \tag{5.1}$$

$$k(\sigma) \in \mathbb{Z}/2\mathbb{Z} \qquad \text{if } \sigma = \sigma c \quad \text{(possible supersingular factor)}. \tag{5.2}$$

In (5.1), $k(\sigma)$ is given by Theorem 3, extended, as in §4, to the case where E is CM rather than imaginary quadratic. Note that if σ is changed to σc, the spaces V_1 and V_2 of §4 are exchanged, which gives relation (5.1).

Let K be the group of functions satisfying (5.1) and (5.2). Since the vector space over $\mathbb{Q}_\lambda$ generated by algebraic cycles is stable under G and under cup product, it corresponds to a submonoid Alg of K. Moreover,

$$\mathrm{Alg} \subset K_0 = \left\{ k : \sum_{\sigma \bmod c} k(\sigma) \text{ is even} \right\}, \tag{5.3}$$

$$H_\lambda(k) \perp H_\lambda(\ell) \quad \text{if } k + \ell \neq 0, \qquad H_\lambda(k) \text{ and } H_\lambda(-k) \text{ are in duality}, \tag{5.4}$$

$$\mathrm{Alg} \text{ is stable under } \mathrm{Gal}(\mathbb{Q}_\lambda/\mathbb{Q}_\ell). \tag{5.5}$$

Assume that the Galois group contains an element acting as -1 on K. Then, on every $\mathbb{Q}_\ell$-rational subspace of H_λ invariant under G, the duality given by the intersection form is non-degenerate. This is in particular the case if, with the notation of §1, complex conjugation $c \in \mathrm{Gal}(E/\mathbb{Q})$ belongs to the decomposition group of ℓ, proving Theorem 1.

Slightly less is enough: in order that every stable submonoid be a subgroup of K (so that the space of algebraic cycles contains, together with the space associated with k, its dual associated with $-k$), it is enough that $K \otimes \mathbb{Q}$ have no fixed line under the action of $\mathrm{Gal}(\mathbb{Q}_\lambda/\mathbb{Q}_\ell)$. In other words, if ℓ is unramified in the factors of E, it is enough that Frob_ℓ, acting on Σ, satisfy:

$$\text{for every } \sigma \in \Sigma, \text{ there is } a \in \mathbb{Z} \text{ such that } \mathrm{Frob}_\ell^a(\sigma) = \sigma c.$$

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